

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

28th October 2015

Subject CT6 – Statistical Methods

Time allowed: Three Hours (10.30 – 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions, beginning your answer to each question on a separate sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** An actuarial student travels by local train to his music coaching classes which are held for a maximum of 5 days a week. However the classes are held at random and he is informed one day prior if classes are scheduled for the next day. His daily trip cost one way to his destination is Rs 50. He needs to decide if he wishes to purchase a weekly train pass which will cost him Rs 320. The probability distribution of the no of times coaching classes are held in a week are:

Number of classes held in a week	Probability
1	.2
2	.1
3	.3
4	.25
5	.15

Given that he never misses a scheduled class and travels by train only, find out his minimax and Bayes criterion decisions.

[5]

- Q. 2)** Five Pseudo-random numbers (3 digit) are generated as below:

271, 399, 240, 704, 285

Using the above numbers generate a random sample from Chi-square distribution with 10 degrees of freedom.

[7]

- Q. 3)** The number of claims per policy per year follows Poisson distribution with unknown parameter μ . Prior beliefs that μ follows Gamma distribution with parameters α and λ . Number of policies sold, individual claim amounts and average total claim amount per policy is summarized in the table given below.

Item	Year 1	Year 2	Year 3	Year 4
No of policy	100	130	180	250
Individual claim amounts	1,500	1,700	2,000	2,500
Average total claim amount per policy	a1	a2	a3	A4

- i) Using posterior distribution of μ , derive $E(A4 | a1, a2, a3)$ in a credibility estimate form as given below:

$$2500 \times \frac{\alpha}{\lambda} \times (1 - Z) + b \times Z, \text{ specifying } b \text{ and } Z.$$

(7)

- ii) Comment on Z.

(1)

[8]

Q. 4) An Insurer writes two classes of insurance business A and B.

- i) On class A the insurer is planning to buy a reinsurance policy such that the reinsurer pays the excess of the claim amount over 250, if the claim amount exceeds 250. However, in case the claim amount exceeds a pre-defined amount M (which is greater than 250), the excess claim amount over M reverts back to the insurer. No payment is made by the reinsurer if claim payments are less than or equal to 250.

Assuming that the original claim payments follow an exponential distribution with mean 1000 calculate M such that the expected claim payments made by the reinsurer equals 500. (8)

- ii) On class B the insurer decides to introduce a policy excess such that 40% of the losses result in no claim for the insurer. The insurer believes that claims from class B follow a Pareto distribution with parameters $\alpha = 3$ and $\lambda = 600$. (7)

Calculate the average claim amount paid by the insurer under class B net of the policy excess.

[15]

Q. 5) An Insurance Company operates in three different regions namely North, South & West and the past claim experience for these regions are summarized in the table given below.

<i>Figs are in Crs</i>				
Accident Year	Region	Development year		
		0	1	2
FY 13	North	50	40	65
	South	30	55	50
	West	55	60	35
FY 14	North	45	60	
	South	55	50	
	West	40	55	
FY 15	North	60		
	South	65		
	West	50		

- i) Calculate ultimate claim amount for each accident year and for each regions using chain-ladder method. (6)

The number of in-force policies for different regions and different accidental year are given below:

<i>Figs are in lacs</i>			
Region	No of policy		
	FY 13	FY 14	FY 15
North	25	32	30
South	30	34	28
West	27	26	38

The expected number of in-force policy for South region for the year FY 16 and FY 17 are 35 lacs and 40 lacs respectively.

- ii) Calculate expected ultimate claim amount for South region for accident year FY 16 and FY 17 using Empirical Bayes Credibility theory Model 2. (9)
- iii) State the assumptions under the above model. (1)
- iv) For south region, calculate the expected reported claim in reporting years FY 16, FY 17, FY 18 and FY 19 (separately) for claims incurred till accident year FY 17. (4)

[20]

Q. 6) A motor insurance company is selling insurance for car, truck, two wheelers and buses. The total numbers of years of experience for each type are 5, 3, 10 and 6 years respectively. The number of claims in any given year and for any given risk is assumed to follow a Poisson distribution. The total number of claims for each type till date is 65, 45, 85 and 75.

- i) Calculate maximum likelihood estimate for each insurance type. (4)
- ii) Perform a test using scaled deviance test, to check whether mean claim rates are different for each insurance type. (5)

[9]

Q. 7) On the occasion of the festival Deepawali, a local shop ChocoRich announces a contest for its loyal customers. It makes two varieties of chocolate boxes. Each box contains 10 chocolates. The first variety "Raisin Rich" contains 3 nut chocolates and 7 raisin chocolates while the second variety "Nut Rich" contains 8 nut and 2 raisin chocolates. Both the raisin and nut chocolates are similar in shape and size and can only be identified by experiencing the taste. The price of each box is Rs 100.

A customer is given one opportunity to taste any one chocolate from a randomly chosen box. If he guesses the box correctly then ChocoRich offers a discount of 25% else a discount of 10% is offered.

An actuarial student happens to be a loyal customer of the shop and decides to take part in the contest.

- i) Calculate the value of the risk function for each decision function and the two chocolate boxes. (8)
- ii) Under the Bayes Criterion find out the optimum strategy used by the actuarial student if out of every 10 chocolate boxes produced by ChocoRich, 4 are of the variety "Raisin Rich". (3)

[11]

- Q. 8)**
- i) Define the two commonly used loss functions apart from quadratic loss function. (2)
 - ii) Show that the Bayesian Estimate under quadratic loss is the mean of the posterior distribution. (3)

[5]

Q. 9) The number of accidents (N) made by any individual vehicle over a period of 5 years is assumed to follow the density function as :

$$P(N = 0) = 0.75$$

$$P(N = 1) = 0.15$$

$$P(N = 2) = 0.08$$

$$P(N = 3) = 0.02$$

$$P(N > 3) = 0.$$

i) Describe carefully how you would use the acceptance/rejection method to generate discrete pseudo-random values from this distribution using as your reference distribution the discrete uniform distribution that is equally likely to take any of the values 0, 1, 2 or 3. (You are not required to describe how the uniform values are generated and you may assume that you have continuous uniform values available for the acceptance/rejection decisions.) (4)

ii) What proportion of values generated by the method in (i) will be accepted?

Comment on your answer. (3)

iii) Outline a more efficient alternative method that could be used to generate the pseudo-random numbers for the numbers of claims. (3)
[10]

Q. 10) i) Define “Adjustment Coefficient” of a compound Poisson process. (1)

ii) An insurer calculates the annual premiums for group fixed benefit (major and minor) one year micro health insurance policies by increasing the risk premium by 30% and adding a Rs 30 per policy loading. The probability claim is 3% and individual claim amounts can be assumed to be Rs. 2000 with probability 0.9 (minor) and Rs 15000 with probability 0.1 for major benefit. Only one claim (either major or minor) per life can be made over one year.

Calculate the insurer’s adjustment coefficient for these policies, to 2 significant figures (9)
[10]
